

Geometry

Unit 10

Lesson 1 Homework

Name _____ Answer Key _____

Date _____

Period _____

Use the given context to answer questions 1-10.

Jamar orders an extra-large circular pizza that has an 18-inch (in.) diameter and is cut into 12 triangular slices.

1. What is the approximate area of an 18-inch pizza? Use 3.14 for π .

$$A = 3.14(9)^2 = 3.14(81) = 254.34 \text{ in}^2$$

2. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 6 slices?

$$\text{Fraction: } \frac{1}{2}$$

$$\text{Percentage: } 50\%$$

3. What is the area of 6 slices?

$$A = 127.17 \text{ in}^2$$

4. What is the central angle of 6 slices?

$$180^\circ$$

5. If the slices are cut precisely and equally, what fraction of the pizza is 8 slices?

$$\frac{8}{12} = \frac{2}{3}$$

6. What is the area of 8 slices?

$$A = \frac{2}{3}(254.34) = 169.56 \text{ in}^2$$

7. What is the central angle of 8 slices?

$$240^\circ$$

8. If the slices are cut precisely and equally, what fraction of the pizza is 1 slice?

$$\frac{1}{12}$$

9. What is the area of 1 slice?

$$A = \frac{1}{12}(254.34) = 21.195 \text{ in}^2$$

10. What is the central angle of 1 slice?

$$30^\circ$$